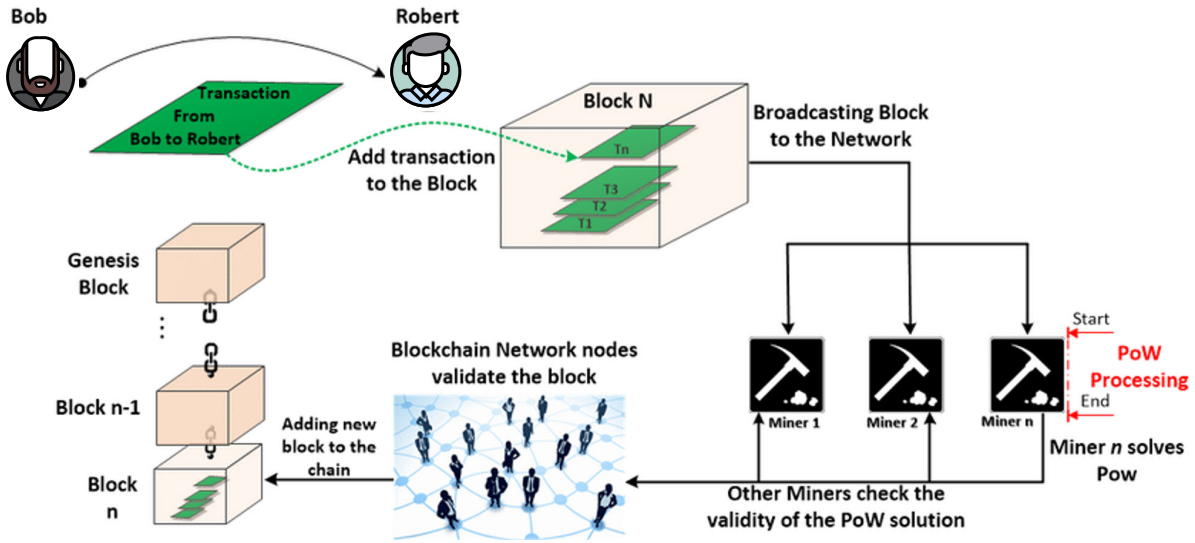
8.5.2 Summary of the Transaction from Start to Finish

1. A user wants to send some ***bitcoin*** to another user. They create a transaction with the details of the send, including the amount of ***bitcoin*** being sent, the sender's address, and the receiver's address.
2. The user then uses their **private key** to encrypt the transaction. This private key is like a secret code that only the user knows and it is used to prove that the user is who they say they are.
3. The encrypted transaction is broadcasted to the network of ***Bitcoin*** nodes.



4. The nodes verify the transaction using the sender's **public key**, which is available on the *blockchain*. They check that the signature is valid and that the sender has enough ***bitcoin*** to complete the transaction.
5. The nodes then group the verified transactions into a block.
6. The block is then broadcasted to the network of ***Bitcoin*** miners.
7. Miners use a complex mathematical algorithm to solve a puzzle, which is called "mining." Once the puzzle is solved, it is added to the *blockchain* and the block is added to the chain.
8. Once the block is added to the *blockchain*, the transaction is considered complete and the receiver can access the ***bitcoin*** using their own **private key**.

In summary, the sender creates and encrypts the transaction with their private key, the nodes verify the transaction UTXOs using the sender's public key, and the miners add the verified transaction to the blockchain. The receiver can then access the ***bitcoin*** using their private key. Once a block is mined, all the transactions included in it are considered confirmed, and the UTXOs used as inputs in these transactions are considered spent and will not be used again.